20/2/24

CS23532 - COMPUTER NETWORKS LABORATORY

EXPT-1: BASIC NETWORKING COMMANDS

AIM:

Study of various network commands used in Linux and Windows commands.

COMMANDS:

i) arp -a: ARP is a short form of address resolution protocol which show IP address of your command along with IP address and MAC address of your router.

OUTPUT:

Interface 172.16.893 --- 0x4

Internet Address	Physical address	Type
172.16.8.1	7c-5a-1c-c6-be-45	dynamic
172.16.8.172	7c-57-58-35-0d-3d	dynamic
172.16.11.255	ff-ff-ff-ff-ff-ff	static
244.0.0.2	01-00-5e-00-00-02	static
244.0.0.251	01-00-5e-00-00-fb	static
239.255.255.250	01-00-5e-7f-ff-fa	static.

ii) Hostname:

This is simplest of all TCP/IP commands. It simply displays name of your computer.

OUTPUT:
localhost - local domain

iii) Ipconfig/all:

This command displays detailed configuration information about your TCP/IP connection including router, gateway, DNS, DHCP and type of ethernet adapter in your system.

OUTPUT: Windows IP configuration

Hostname : DESKTOP-V4FUKL3
Primary Dns Suffix :
Node type : Mixed
IP Routing Enabled : No
WNS Proxy Enabled : No

iv.) nbtstat -a:

- This command helps solve problems with NETBIOS name resolution.

(Nbt → NetBIOS over TCP/IP).

v.) Netstat: [Network statistics]

It displays a variety of statistics about computers active TCP/IP connections. It is a command line tool for monitoring network connections both incoming and outgoing as well as routing table.

OUTPUT:

Proto	Recv-Q	Send-Q	L-Addr.	F.A	State
Proto	Refcnt	Flags	Type	State	I-Node
unix	2	[ ]	DGRAM	=	28018

vi) nslookup:

[name server lookup] is a tool used to perform DNS lookups in ~~Unix~~. It is used to display DNS details such as the IP address of particular computer.

OUTPUT:

Server: 172.16.8.1

Address: 172.16.8.1.53

vii) Pathping:

Pathping is unique to window's and is basically combination of ping and Tracert commands.

viii) Ping:

Packet Internet groper command is best way to ~~test~~ connectivity b/w two nodes. Ping use ICMP to communicate to other devices.

OUTPUT:

PING localhost (localhost (::1)) 56 databytes

ix.) Route:

It is used to show manipulate the IP routing table. It is used to setup static routes.

OUTPUT:

Kernel IP routing table

Destination	Gateway	Genmask	Flags	Metric	Ref
10.0.2.0	0.0.0.0	155.255.255.0	U	100	0

LINUX COMMANDS

1.) ip <OPTIONS><OBJECT><COMMAND>

ip address show

ip address add 192.168.1.254 dev enp

ip address del 192.168.1.254/24 dev enph

ip route add default via
192.168.1.254 dev etho

2.) ifconfig

3.) mtr (matt's trace route)

mtr <options> hostname / IP

mtr google.com

mtr -g google.com

mtr -c 10

4.) tcpdump

dnf install -y tcpdump

tcpdump -i etho -c 10

tcpdump -i etho src host 8.8.8.8

tcpdump -i etho -c 10 host
www.google.com and port 443

5.) Ping

Ping google.com

Ping -c 10 google.com

Configuring an Ethernet connection by using nmcli.

1.) nmcli connection show

NAME	UUID	TYPE
Wired connection 1	95ebf6490-cc-20-366-8	ethernet

2.) nmcli connection add con-name <connection name> ifname <device-name> type ethernet.

3.) nmcli connection modify "Wired connection".

4.) nmcli connection show

connection interface name: enp1s0

connection.autoconnect : yes

ipv4 method : auto

ipv6 method : auto.

5.) ~~Configure~~ IPV4

nmcli connection modify "Wired connection 1" ipv4.method auto.

6.) # nmcli connection up Internet - LAN

RESULT:

Thus the various network commands used in Linux and Windows are studied and output was verified.

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EXP. NO: 2 : STUDY OF NETWORK CABLES

AIM:

To study of different types of Network cables.

a.) Understand different types of network cable

Different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) cable
3. Coaxial Cable
4. Fibre Optic cable.

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use.
UTP	Category 3	10 bps	Advantages: * cheaper in cost. * Easy to install as they have smaller overall diameter. Disadvantages: * More prone to (EMI) interference & noise.	10 Base-T Ethernet
	category 5	up to 100 Mbps		Fast ethernet, Gigabit ethernet
	category 5e	1 Gbps		Fast Ethernet, Gigabit Ethernet
STP	Category 6, 6a	10 Gbps	Advantages: * Shielded * Faster than UTP * Less susceptible to noise	• Gigabit ethernet. 10G ethernet (55 m) • used data centres

SSTP	Category 7	100Gbps	Disadvantages: * Expensive * Greater than installation effort	Gigabit Ethernet (a Ethernet (100 m)
Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	Advantages: * High bandwidth * Immune to interference * Low-Loss bandwidth * Versatile Disadvantage: * Limited distance. * cost * Size is bulky	Speed of signal is 500m. Television network High speed Internet connections.
Fibre optics cable	Single mode Multi mode	100Gbps	Advantages: * High speed * High bandwidth * Long distance Disadvantages. * Expensive * Required Skilled Installers	Maximum distance of fibre optics cable is around 100 meters.

Result:

Thus the study of different types of network cables is done successfully.

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EXP-3: Experiments on cisco Packet tracer (simulation Tool)

AIM:

To study the Packet tracer tool Installation and User Interface overview

INTRODUCTION:

Cisco Packet Tracer is a network simulation tool used to create, configure and troubleshoot computer networks without using real hardware.

It helps students understand the working of network devices such as PCs, Hubs, Switches and Routers.

INSTALLATION:

1. Download Cisco Packet Tracer from the cisco Networking Academy website.
2. Log in with your cisco account
3. Download the version suitable for your operating system.
4. Run installer and follow installation instructions.

PACKET TRACER UI

The main parts of Packet Tracer UI are:

1. Menu Bar - used for opening, saving and managing files.
2. Main Toolbar - contains shortcuts such as save, open, undo, redo & zoom.
3. Logical / Physical Workspace - used to switch between logical and physical views.
4. Workspace - Area where network topology is created.
5. common Tools Bar - Contains tools for selecting, deleting, inspecting and adding Packets.
6. Realtime / Simulation Mode - Used to view network normally or observe packet movement
7. Network Component Box - Contains routers, switches, hubs, PCs.
8. Packet Box - used to create and observe test packets.

PROCEDURE:

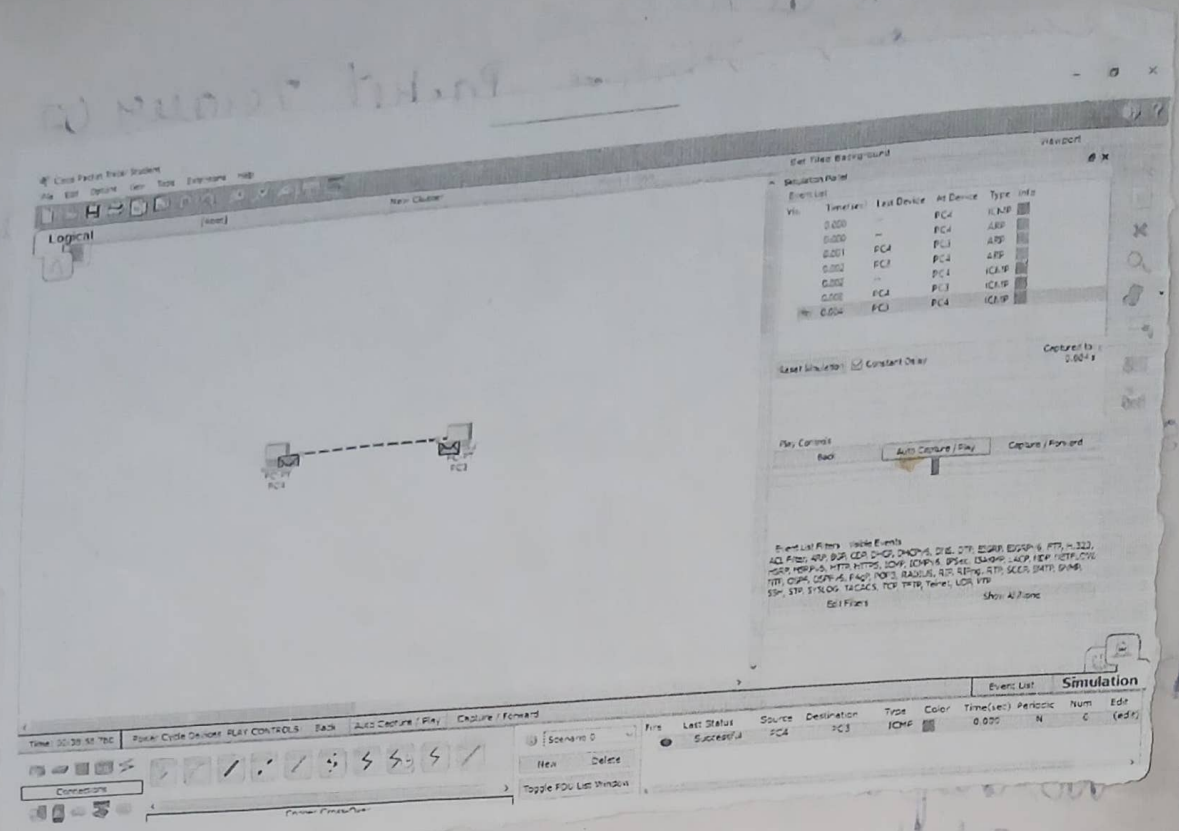
1) Peer-to-Peer Network

Open Cisco Packet Tracer.

Add 2 PCs to the workspace.

Select connections and choose copper cross over cable.

Connect PC0 directly to PC1

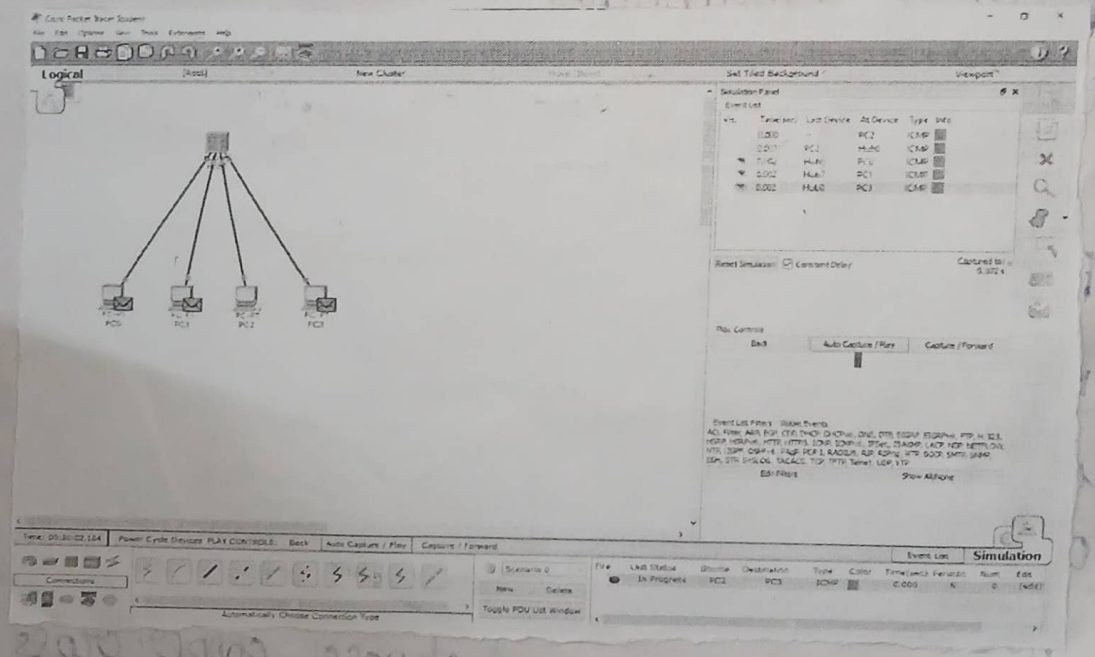


2) Network using Hub

i) Open Cisco Packet Tracer.

ii) Add 4 Pcs and 1 Hub to workspace.

iii) Connect all Pcs to hub using copper straight through cables.



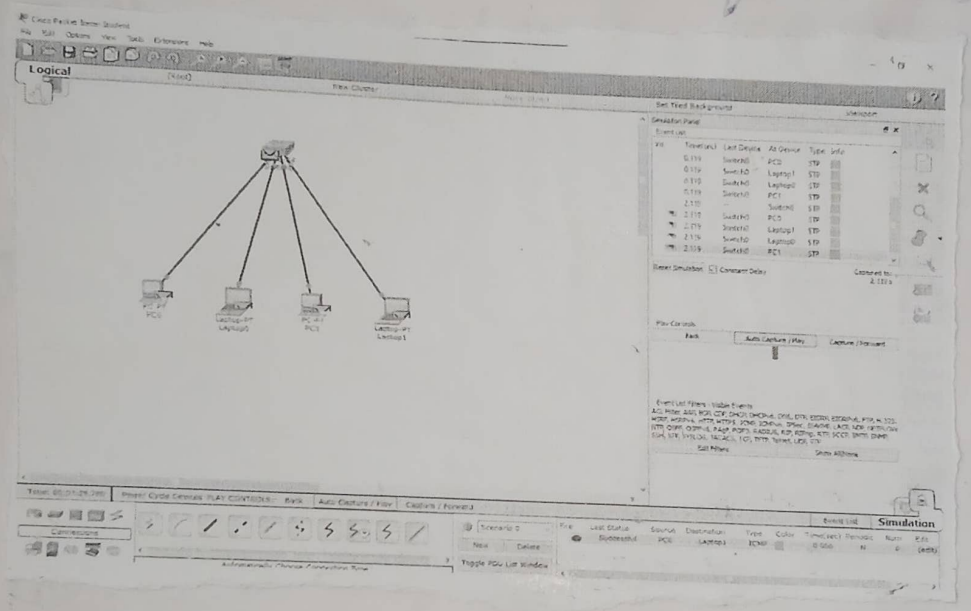
3.) Network using Switch

Add 4 PCs and 1 Switch.

connect all PCs to switch using copper straight through cables.

configure IP addresses in same network.

Send a PDU from one PC to another.



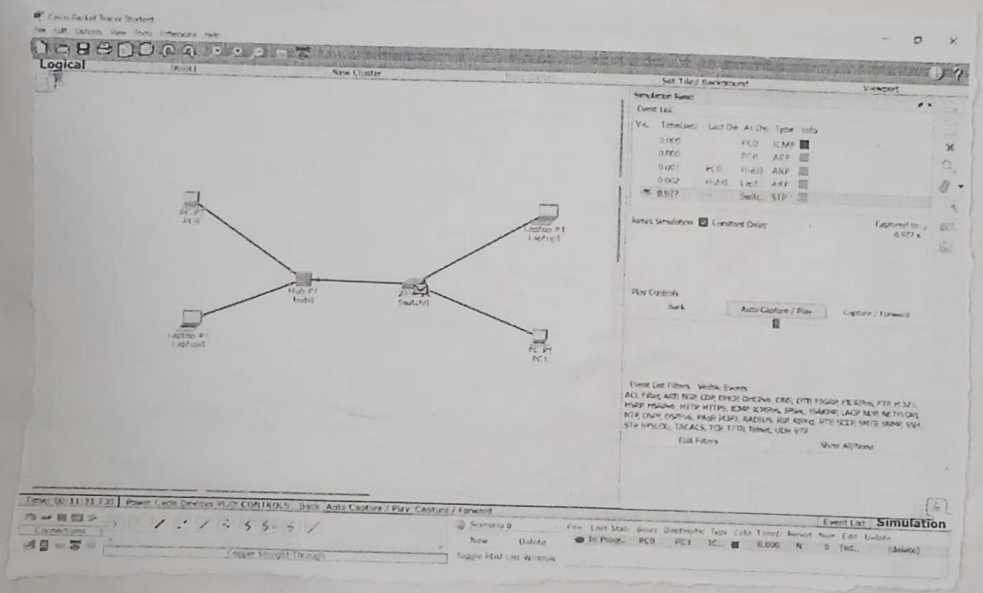
1.) Network using Both Hub and Switch.

Open Cisco Packet Tracer.

Add 4 PCs, 1 Hub and 1 Switch to workspace.

Connect PC0 and PC1 to Hub using copper straight through cables.

Connect PC2 and PC3 to switch using copper straight through cable.



RESULT:

Thus the Study of Packet Tracer tool is Implemented successfully.

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Ex.NO: 4
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Setup and configure a LAN using a switch and Ethernet cables.

AIM:

To setup and configure a LAN (Local Area Network) using Switch and Ethernet cables.

What is LAN?

A LAN is a network that connects computers and other devices within a limited area such as home, school or office. It allows device to communicate and share resources such as files and printers.

REQUIREMENTS:

4 computers
1 Network switch
4 Ethernet cables.

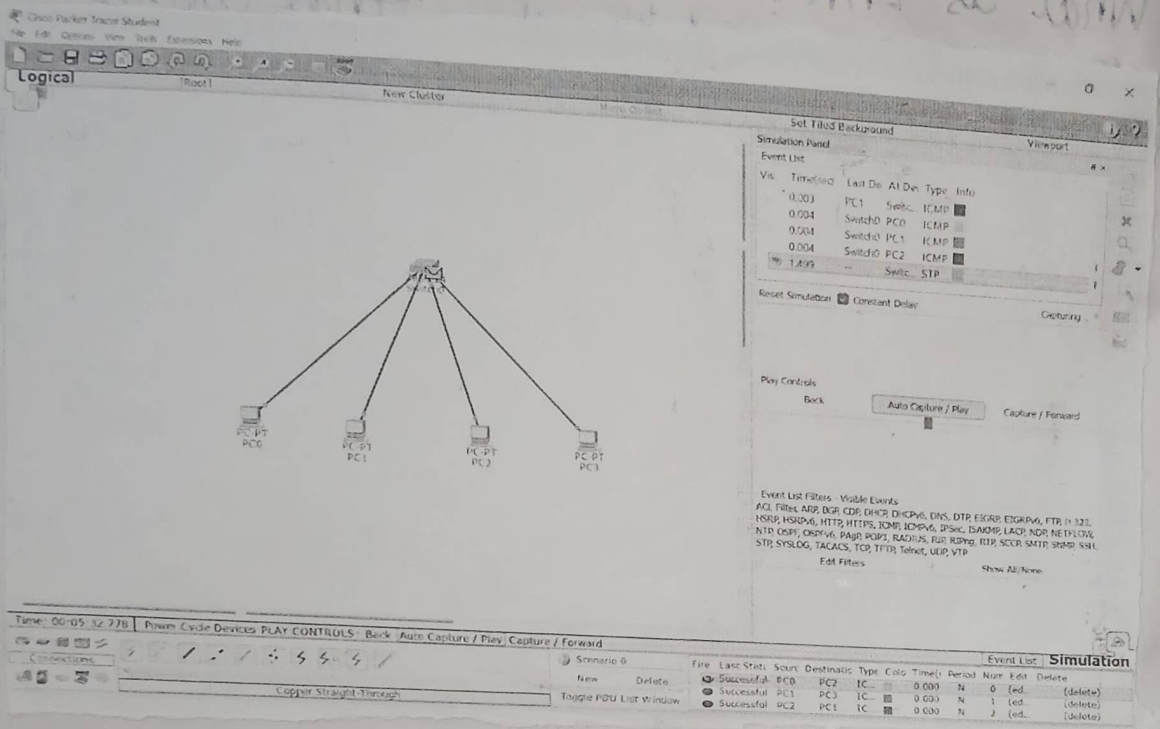
PROCEDURE:

1. connect 4 computers to switch using ethernet cables.
2. Assign IP addresses to the computers.
   PC1: 10.10.10.1
   PC2: 10.10.10.2
   PC3: 10.10.10.3
   PC4: 10.10.10.4
3. Configure switch
4. Open command prompt on a computer and use Ping command to check connectivity.

5. Select a folder on one computer and enable file sharing for users on same LAN.

6. Access shared folder from other computers on network.

7. Verify that all computers can communicate and share files sucessfully.



RESULT:

Thus to setup LAN using Switch and Ethernet is created sucessfully.

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Ex.no: 5 PACKET CAPTURE TOOL: WIRESHARK
18/8/26

AIM:

To understand the features of wireshark as a packet capture tool and understand encapsulation of information at various layers of a protocol.

PROCEDURE:

Step1: Open wireshark and select wifi network interface.

Step2: Start the packet capture by double clicking the wifi interface.

Step3: Observe the packets being capture in wireshark window.

step4: Different packets such as ARP, DHCP, NBNS observed in packet list.

step5: Select any packet to view its detailed information such as frame, ethernet, IP, UDP/TCP

step6: Apply display bar to display only DNS packets.

step7: Observe DNS query and packets the source and destination IP address, protocol, packet length.

step8: Stop packet capture after collecting required packets.

RESULT:

Thus, Network packets work Successfully captured and analysed using Wireshark. and DNS packets over identified using display filter.

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Ex:6 : ERROR CORRECTION AT DLL

21/8/26

AIM:

Write a program to implement error detection and correction using Hamming code concept.

CODE:

```
data = input("Enter a data bits:")
d1, d2, d3, d4 = map(int, data)
P1 = d1 ^ d2 ^ d4
P2 = d1 ^ d3 ^ d4
P4 = d2 ^ d3 ^ d4
code = [P1, P2, P1, P4, d2, d3, d4]
print("Hamming code:", .join(map(str, code)]
received = set(map(int, input("Enter 2nd code:"))
S1 = received[0] ^ received[2] ^ received[4]
                                ^ received[6]
S2 = received[1] ^ received[2] ^ ruived
                    [5] ^ received[6]
S4 = received[3] ^ received[4] ^ received[5]
                    ^ received[6]

error = S4 * 4 + S2 * 2 + S1
if error = 0,
    print("No error")
else
    print("Error at position:", error)
received[error-1]^ = 1
```

Print (corrected code: ", ". join(map(str, received))).

O/P:

Enter 4 data bits: 1001
Hamming code: 0011001
Enter received code: 1001011
Error at position: 4
Corrected code: 1000011

RESULT:

Thus, hamming code was successfully generated, error was detected and bit was correctly placed.

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Ex:7 : FLOW CONTROL AT DATA LINK
6/9/26

AIM:

write a program to implement flow control at data link layer using sliding window protocol, simulates flow of frames from one node to another.

CODE:

sender.py

```
message = input("Enter a message: ")
window = int(input("Enter window size: "))
i = 0
while i < len(message):
    print("\n Sender Buffer: ")
    for j in range(i, min(i+window, len
                    (message))):
        print("frame", j+1, ": ", message[j])
    ack = int(input("Enter ACK received: "))
    if ack >= i+1 and ack <= min(i+window,
                    len(message)):
        print("ACK accepted")
        i = ack
    else:
        print("Wrong ACK - Resending frame")
print("\n All frames sent
                    Successfully")
```

Receiver.py:

```
frame = int(input("Enter received frame
                   number: "))
data = input("Enter frame data: ")
expected = int(input("Enter expected
                      frame number: "))
print("\n Receiver")
if frame == expected:
    print("Frame", frame, "received")
    print("Data:", data)
    print("ACK", frame, "sent")
else:
    print("Wrong frame")
    print("NACK sent")
```

O/P:

Sender:

```
Enter message: hello
Enter windowSize: 3
Sender Buffer:
Frame 1: h
Frame 2: e
```

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Exp.No:8 NMAP to discover Live Hosts using
08/9/26 Nmap scans on Try HackMe Platform

AIM:

To discover Live Hosts using Nmap scans (ARP, ICMP, TCP/UDP) on Try HackMe Platform.

NMAP:

Nmap stands for Network Mapper. It is an open-source network subnetting scanning tool used to identify live hosts & network services.

send a packet with following

From:
computer 1

To
computer 1

Packet Type
arp_request

Data
computer 6.

From computer 1
To computer 1
Packet Type: ARP Request

1.) How many devices can see ARP request?

2) Did computers receive ARP Request ?
may be

Send a packet with following

From
computer 4
To
computer 4
PacketType
arp-request
Data
computer, 6

From computer 4
To computer 4
Packet Type: "ARP Request"
Data: computer 6.

1.) How many devices can see ARP request? 4
2.) Did computers reply to ARP request? yes

send a packet
From pc0 computer 1
To computer 3 Packet type: "Ping Request"

1) What type of packet did computer 1 send before Ping? ARP request.
2) How many computers responded to ping request ?
1

RESULT:

Thus to discover Live Hosts using N-map scans on TryHackMe Platform is executed successfully.

Ex. No: 9
20/9/26

Implementation of Subnetting in cisco Packet Tracer Simulator.

AIM:

To implement subnetting in cisco Packet Tracer Simulator.

Creating a network topology.

The first step in implementing classless IP subnetting is to create a network topology in Packet Tracer. Select "New" button in top left corner

Subnetting:

To subnet the network address of 192.168.1.0 to provide enough space for at least 5 addresses for end devices, the switch and router, we can use a /27 subnets with 30 host addresses each.

RESULT:

Thus an experiment to implement subnetting in Cisco Packet Tracer is executed successfully.

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Ex.No: 10 PRACTICAL 10(a)
13/11/26

AIM:

To simulate routing information Protocol (RIP) using Cisco Packet Tracer and observe dynamic routing b/w diff. networks.

IP configuration:

Device	Pco	Interface fast ethernet	IP address
Router 0		Po0/1	10.0.0.2/8
Router 1		s0/0/1	10.0.0.1/8
Router 1		S0/01,	192.168.1.254/
Router 2		S0/0/0	192.168.1.246/3
PC1		fast ethernet	192.168.1.253/30

PROCEDURE:

1.) Create topology with 3 routers and 2 PCs

2.) Connect routers using serial Cables.

3.) Assign IP addresses to all PCs and routed interface.

4.) Enable router interface.

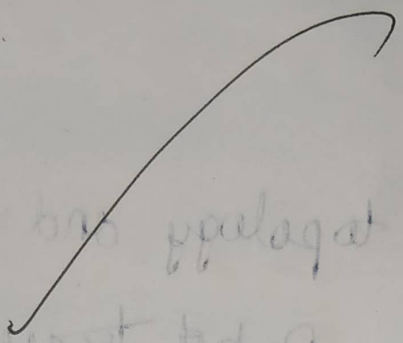
5.) configure remaining interface.

6.) Enable RIP on router 0 and

configure RIP on router1 and router 2.

7.) From PC0, CISC ping 20.0.0.2 to verify connectivity.

8.) We trace to observe route taken by packets.



RESULT:

Thus RIP was successfully simulated in CISCO Packet Tracer and dynamic routes select and routing were observed.

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13/10/26 Practical 10 (b)

AIM:

To design and configure an internetwork using wireless router, DHCP server and cloud.

PROCEDURE:

Part 1: Build topology and connections.

1. Launch the Packet tracer and place Network devices.

2. Connect the devices ~~using~~ physical cables.
   → PC to wireless Router
   → Wireless router to cable modem.

Part 2: Configure Network devices.

1. Wireless Router configuration
   → open wireless router on 0, navigate to LAN1
   → Go to setup tab
2. Laptop configuration
3. Setup internet cloud configuration

Part 3: Verify connectivity & commands

1. Open command prompt on PC to verify and renew network configuration using commands.

Cmd:

ipconfig/all

ipconfig/release

ipconfig/renew.

RESULT:

Thus Internetwork was successfully designed and configured using wireless router DHCP server.

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Practical #12

AIM:

To simulate static routing configuration using cisco packet tracer.

Procedure:

1. Configure primary router with lower administrative distance.
2. Routers automatically learn connected networks static routes must be manually added..
3. It allows router to automatically slect primary path.

Commands:

Router > enable

Router # configure terminal

Router (config) # ip route 30.0.0.0 25.0.0.20

Router (config) # ip route 30.0.0.100.255.0 255.25

O/P:

30.0.0.0.018 is variable subnetted.

S 30.0.0.018 [10/0] via 20.0.0.2

S 30.0.0.100/32 [10/0] via 40.0.0.2

2. Router configuration:

Router > enable
Router # configure terminal
Router (config) # ip route 10.0.0.0 255.0.0.0 20.0.0.1
Router (config) # ip route 10.0.0.0 255.0.0.0 50.0.0.01
Router (config) # exit

O/P:

S 10.0.0.0/8 [10/0] via 20.0.0.1
S 40.0.0.0/8 [10/0] via 20.0.0.1

4. Verification and Path tracing O/P:

cmd:

C:\> tracert 30.0.0.20

O/P:

Tracing route 30.0.0.20 : Maximum of 30 hops

1	0ms	0ms	10.0.0.1
2	0ms	0ms	20.0.0.2
3	*	0ms	30.0.0.20

Result:

Thus static routes were successfully configured.

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PRACTICAL 13

AIM.

Implement echo client server using TCP/IP sockets.

CLIENT CODE:

```
import socket
HOST = "127.0.0.1"
PORT = 5000
client = socket.socket(socket.AF_INET,
        socket.SOCK_STREAM).
client.connect((HOST, PORT))
while TRUE:
    message = input();
    client.send (message.encode())
    if Message.lower() == "exit";
        break
    data = client.recv(1024).decode()
    client.close().
```

SERVER CODE:

```
import socket
HOST = 127.0.0.1
```

```
PORT = 5000
Server = socket.socket(Socket.AF_INET,
                       Sock)
Server.bind(HOST, PORT)
server.listen()
com, addr = Server.accept()
While TRUE:
    data = conn.recv(1024).decode()
    if not data or data.lower() == "exit":
conn.send(data.encode())
conn.close()
server.close()
```

RESULT:

Thus code has been executed ..

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4/10/26 PRACTICAL 14:

AIM:

To write a code using 'RAW' sockets to implement packet sniffing.

CODE:

```
from scapy.all import sniff
from scapy.layers.inet import IP.
def packet_call back(packet):
    if IP in packet:
        ip_layer = Packet[IP]
        Protocol = ip_layer.proto
        Size_ip = ip_layer.src
        det_ip = ip_layer.dist
    if protocol == 1:
        protocol_name = ICMP
    elif protocol == 6:
        protocol_name = TCP
    else
        protocol_name = "unknown protocol
    Print("Protocol:", protocol_name)
    Print("-" * 50)
```

Security

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RESULT:

Thus, packet sniffing is implement

15/10/20

PRACTICAL-15:

AIM:

To analyze different types of web logs using webaliser tool.

PROCEDURE:

Step 1: Run Webaliser Window Version

Step 2: Input weblog file (download from web)

Step 3: Press Run Webaliser.

O/P:

15. Webalizer (Server Analysis)